28/3/26

JS - test

1) JavaScript & use?

JavaScript is a programming language used to make web pages interactive and dynamic.

Why we use it:

* To handle user actions (clicks, input)
* To update content without reloading
* To build webapps.

2) Variables:

A variable is used to store data values

Types:

* var
* let
* const

Example:

```
var userName = "React";
let todayTest = "JS";
const submit_on = "git";
```

3) Data types

Data types define the type of value

* String → "name"
* Number → 1, 2
* Boolean → true or false

4) D/B let, var and const

Feature	var	let	const
Scope	Function	Block	Block
Reassign	Yes	Yes	No
Redeclare	Yes	No	No

Example :

```
let x = 10;
x = 20; //allowed
```

```
const y = 5;
y = 10; //not allowed
```

5) Operators :

Operators perform operations on values

Types :

1) Arithmetic (+ - * /)
2) Assignment (=)
3) Logical (&& ||)
4) Comparison (== ><)

6) D/B == and ===

== → Compares value only

=== → Compares value + type

Example :

```
Console.log (5 == "5"); // true
Console.log (5 === "5"); // false
```

7) Function & why use it?

A function is a block of code that performs a task.

Why use:

* Reuse code
* Reduce repetition

Example :

```
function greet() {
    Console.log ("Hello!");
}
greet();
```

8) Function to add two numbers :

```
function add (Num1 , Num2) {
    return Num1 + Num2;
}
Console.log (add (5, 3));
```

o/p

8

9) Array

* Array is a collection of data types
* One variable we can store multiple values
* use : square bracket [ ]
* Each value has an index values

Examples.

```
let fruitName = ["papaya", "apple", "orange"];
```

10) Access array elements?

Arrays are access by index values

```
let fruitName = ["papaya", "apple", "orange"];
console.log (fruitName [0]);
console.log (fruitName[1]);
```

11) Object:

Object is a collection of key and value pairs

key & values are separated by using colon [:]

```
LHS : RHS
key : value
{name : "Is"}
```

13) Array of objects :

→ It is an combination of both array & objects

→ To access this only by for loop or array method.

```
let studentsList = [ {name : "Ram", age : 20},
                     {name : "sam", age : 22},];
```

14] Loop to print names from array of objects :

```
let studentList = [ {name : "Ram"} , {name : "sam"},
                    { name : "sohu"} ];

for ( let i = 0; i < studentList . length ; i++ ){
        console.log (student [i] . name);
}
```

15)

Array	Object
structure : ordered	structure : key - value
Access : index	Access : key
Example : ["a", "b"]	Example : {name : "Ram"}
code : let arr = [" 1", "2"];	code : let obj = { name : "apple", color : "red" };